No Regrets: Design regret in clustered wind farms: how wind rose directionality and optimization thoroughness shape the cost of ignoring neighbors

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Abstract. We study *design regret* in clustered wind farms: the AEP gap between a layout optimized with knowledge of neighboring-farm wakes and one optimized in isolation, evaluated under the same neighbors. Our main contribution is methodological. Computing regret requires differencing two large, noisy AEP-optimization outcomes; this catastrophic cancellation amplifies optimizer noise into the difference. We show that regret estimates do not stabilize until $K \geq 300$ multistart restarts—one to two orders of magnitude beyond standard practice—and that any inter-farm coordination study computing AEP differences between strategies inherits this requirement. With this convergence verified, we map design regret across the elliptical wind rose space of Hart (2025). We bound regret two ways: greedy adversarial individual-turbine placement (upper bound, 0.4–2.5 % of AEP across wind climates, dominated by wind rose folding) and regret cross-sections with an identical neighboring farm (realistic scenario, 0.1–0.4 % for the multi-directional roses representative of operational North Sea sites). Both quantities are upper bounds for real planning purposes: the adversarial bound assumes a malicious neighbor; the realistic bound assumes the neighbor copies your design. A robustness check with the TurboPark wake model preserves the qualitative ordering. For policy, the dominant finding is that regret depends strongly on wind rose directionality, so wind-rose-conditional buffer standards could allow denser development at low-risk (bidirectional) sites without compromising high-risk (unidirectional) ones.

1 Introduction

The North Sea is entering an era of unprecedented wind farm clustering. Denmark's Vindmølleparken energy island will connect up to 10 GW across multiple farms in a shared concession zone; the German Bight already hosts over 8 GW with plans for 30 GW by 2050; and the broader North Sea basin is projected to reach 300 GW as part of Europe's 500 GW offshore target. At this density, wind farms no longer operate in isolation. Cluster wake losses in the German Bight are projected to reduce capacity factors by 30 %, with half of the loss attributable to cross-border wake accumulation (Fliegner et al., 2025). The phenomenon—increasingly termed "wind theft" (Finseraas et al., 2024; Du et al., 2026)—has prompted the UK to issue the first national policy guidance on inter-farm wake effects and NYSERDA to recommend minimum inter-farm spacing of 4 nmi (NYSERDA, 2024). Wake-induced velocity deficits have been measured at distances exceeding 50 km (Schneemann et al., 2020; Canadillas et al., 2022), and inter-farm losses are projected to exceed the impact of a century of climate change on North Sea wind resources (Warder and Piggott, 2025).

Yet these emerging policy responses share a common gap: they treat all sites uniformly. Buffer distance recommendations, "good neighbour" design principles, and compensation frameworks do not account for variation in wind climate. This is a significant omission. A site where wind blows predominantly from one direction creates a single, predictable wake corridor that a neighbor-aware layout can avoid. A site with bidirectional wind creates competing corridors that no single rearrangement can escape. The value of coordinated design—and the cost of failing to coordinate—should depend fundamentally on the directional structure of the wind resource.

We formalize this intuition through the concept of *design regret*: the AEP difference between a layout designed with perfect neighbor foresight and one designed in isolation, both evaluated under the same neighbor wakes:

$$R = \mathrm{AEP}(\mathbf{x}^*_{\mathrm{cons}}, \mathbf{n}) - \mathrm{AEP}(\mathbf{x}^*_{\mathrm{lib}}, \mathbf{n}). \tag{1}$$

Design regret isolates the cost of the design decision itself, separate from the total AEP reduction caused by neighbor wakes. By sweeping the elliptical wind rose parameterization of Hart (2025)—which separates directional concentration from bidirectionality—we quantify how wind rose shape governs design regret. The answer is dramatic: regret varies six-fold across the shape space, from 17 GWh/yr under bidirectional conditions to 101 GWh/yr under concentrated unidirectional conditions. Buffer distance requirements inherit this dependence.

Quick et al. (2026) applied this framework to the Danish Energy Island cluster and found negligible design regret for all nine neighbor configurations. We show this is not a general property of offshore wind, but a consequence of DEI's partially bidirectional wind climate ($f \approx 0.38$). Sites in the southern North Sea with strongly unidirectional southwesterly winds sit in the high-regret region of the shape space and are substantially more vulnerable to uncoordinated cluster development.

However, accurately computing these quantities reveals a methodological challenge that undermines all existing regret-like calculations. Layout optimization is highly multimodal (Thomas et al., 2023; Valotta Rodrigues et al., 2024), and regret—as the *difference* between two optimization outcomes—amplifies the effect of local optima. We show that at least 300 restarts are needed for the regret estimate to stabilize—over an order of magnitude more than typical practice.

This paper makes four contributions:

1. A convergence study demonstrating that 300+ multistart optimizations are needed for reliable regret estimation—over an order of magnitude more than typical practice.

2. A systematic sweep of the wind rose shape space showing that design regret varies six-fold with directionality, with concentrated unidirectional sites most vulnerable.

3. Buffer distance analysis quantifying how neighbor separation reduces regret, and how the decay rate depends on wind rose shape.

4. Regret cross-sections with a realistic identical neighbor farm, showing that design regret is an order of magnitude lower than the adversarial upper bound and peaks in the upwind direction.

2 Methods

2.1 Problem formulation

We consider a target wind farm of $N_t = 50$ IEA 15 MW turbines ($D = 240\,m$, hub height 150 m) within a fixed polygonal boundary derived from the Danish Energy Island concession zone. Minimum inter-turbine spacing is $4D$ (960 m). This configuration is representative of near-future North Sea developments: the IEA 15 MW turbine is the reference class for planned GW-scale projects, and the DEI polygon provides a realistic, irregular boundary.

The liberal layout $\mathbf{x}^*_{\text{lib}}$ maximizes AEP assuming no external turbines. The conservative layout $\mathbf{x}^*_{\text{cons}}$ maximizes AEP with specified neighbors $\mathbf{n}$. Both are evaluated with neighbors present to compute regret via Eq. (1). By construction, $R \geq 0$ when both optimizations reach a global optimum, since the conservative search has access to all of the liberal layout's design space plus the additional information about neighbors. With finite multistart budgets neither optimization is guaranteed to be globally optimal, and stochastic noise can occasionally produce $R < 0$. We therefore include the liberal layout itself as one of the candidate starts in the conservative search and define

$$R = \max(\text{best conservative AEP},\ \text{AEP of liberal layout with neighbors}) - \text{AEP of liberal layout with neighbors}. \tag{2}$$

This *regret pooling* step does not bias the result upward in expectation: the conservative search inherits the liberal layout as a free starting point, so the pooled value equals the conservative AEP whenever the search finds anything at least as good as the liberal layout, and falls back to the liberal AEP otherwise—which is the correct answer when the conservative optimum has not been resolved beyond the liberal one. The same trick is needed for any AEP-difference study where $R \geq 0$ holds in theory but optimization is imperfect.

2.2 Wake model

We use the Bastankhah and Porté-Agel (2014) Gaussian deficit model with $k = 0.04$ and root-sum-of-squares superposition. The simulation is implemented in JAX (Bradbury et al., 2018), enabling automatic differentiation through the wake model. AEP is computed over 24 equally-spaced wind direction sectors at a constant wind speed of 9 m/s:

$$\text{AEP} = 8760 \sum_{k=1}^{24} w_k \sum_{i=1}^{N_t} P_i(U_{i,k}), \tag{3}$$

where w_k is the sector frequency, P_i is the turbine power curve, and $U_{i,k}$ is the effective wind speed at turbine i under direction k.

2.3 Wind rose parameterization

Wind rose shape is parameterized using the elliptical model of Hart (2025), which derives the directional frequency distribution from a unit-area ellipse with semi-major axis a (along the prevailing-direction axis) and semi-minor axis $b = 1/(\pi a)$ (perpendicular).

The CDF of a direction θ relative to the prevailing axis is built from the first-quadrant expression (Hart 2025, Eq. 2):

$$F_{Q1}(\theta) = \frac{1}{2\pi} \arctan\!\big(\pi\, a^2 \tan\theta\big), \qquad \theta \in [0, \pi/2), \tag{4}$$

with $F_{Q1}(\pi/2) = 1/4$, and extended to the full circle via four-fold symmetry. Sector probabilities are obtained by differencing the CDF at the sector edges, rotating by the prevailing direction θ_{prev}, and applying a folding multiplier that breaks the front/back symmetry (Hart 2025, Eqs. 3–5):

$$p_i = \underbrace{\big[\tilde{F}(\theta_i + \Delta) - \tilde{F}(\theta_i - \Delta)\big]}_{\text{un-folded elliptical probability}} \cdot \underbrace{\begin{cases} 1 + f, & \theta_i - \theta_{\text{prev}} \in (-\pi/2, \pi/2) \quad \text{(front half)} \\ 1 - f, & \theta_i - \theta_{\text{prev}} \in (\pi/2, 3\pi/2) \quad \text{(back half)} \\ 1, & \theta_i - \theta_{\text{prev}} = \pm\pi/2 \quad \text{(perpendicular to prevailing)} \end{cases}}_{\text{folding multiplier}}, \tag{5}$$

where θ_i is the sector centre, $\Delta = \pi/n_{\text{sec}}$ is the half-sector width, and $\tilde{F}$ is the rotated, periodically-extended CDF $\tilde{F}(\theta) = F(\theta - \theta_{\text{prev}}) + \lfloor(\theta - \theta_{\text{prev}})/(2\pi)\rfloor$. Two shape parameters control the distribution:

- **Shape parameter** $a > 0$: controls the ellipse eccentricity. The special value $a = 1/\sqrt{\pi} \approx 0.564$ gives a circle (uniform distribution). For $a > 1/\sqrt{\pi}$ the ellipse is elongated along the prevailing axis and probability concentrates there (larger $a \Rightarrow$ more concentrated unidirectional rose when $f = 1$). For $a < 1/\sqrt{\pi}$ the ellipse is elongated perpendicular to the prevailing axis, producing a bimodal rose with peaks at $\theta_{\text{prev}} \pm 90°$.
- **Folding parameter** $f \in [0, 1]$: controls front/back symmetry. At $f = 0$ the rose is bidirectional (equal probability from θ_{prev} and $\theta_{\text{prev}} + 180°$); at $f = 1$ all back-half probability is folded onto the front half.

This parameterization separates two physically distinct wind climate properties: how concentrated the energy is along an axis (controlled by a) and whether the rose is symmetric about its prevailing direction (controlled by f). The prevailing direction is fixed at $\theta_{\text{prev}} = 270°$ and wind speed at 9 m/s across 24 equally-spaced sectors. We sweep $a \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ and $f \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ (35 combinations). This range spans the variety of North Sea wind climates, from the strongly unidirectional southwesterlies of the German Bight to the more bidirectional conditions at the Danish Energy Island.

Fitting the DEI wind rose with the elliptical model yields $a \approx 0.64$, $f \approx 0.38$ ($R^2 = 0.84$), placing it in the moderate-concentration, partially-bidirectional regime.

2.4 Adversarial neighbor placement: upper bound on regret

To establish a worst-case upper bound on design regret, we place individual neighbor turbines using a greedy grid search. A candidate grid with $5D$ spacing extends $30D$ from the target boundary, minus a $2D$ exclusion buffer. At each of 30 greedy steps, the single candidate maximizing incremental regret is selected. Evaluating regret at every candidate would require a full K-start conservative optimization per candidate (thousands of SGD solves per step); instead we use a two-stage screen-then-evaluate scheme. *Stage 1 (cheap screening):* for every remaining candidate, we compute the AEP loss that placing a turbine there would induce on the *liberal* layout—a single forward wake simulation, no inner optimization. *Stage 2 (full evaluation):* the top 20 candidates by screening AEP loss receive full K-start conservative re-optimization; the one producing the highest regret is selected and placed. This reduces per-step cost by a factor of 20-25 at the same final result, exploiting the strong correlation between raw AEP loss (which any upwind position induces) and the eventual regret after re-optimization. Figure 1 shows a mid-greedy snapshot with 10 neighbors already placed, the screening AEP-loss field over the candidate grid, and the top 20 candidates that would be fully evaluated at the next step.

This adversarial approach places unconstrained individual turbines—not realistic wind farms—and is designed to find the *maximum achievable regret* for a given wind rose. It provides an upper bound because: (i) individual turbines operate at full thrust coefficient, whereas turbines within a real neighboring farm would experience internal wake losses; and (ii) the greedy algorithm optimizes placement purely to damage the target, whereas a real developer would optimize for their own production. The results from this analysis should be interpreted as the ceiling on design regret, not as a prediction of what a realistic neighbor would cause.

2.5 Regret cross-sections: realistic neighbor scenario

To assess design regret under more realistic conditions, we introduce the *regret cross-section*. An identical copy of the target farm—with the same number of turbines, boundary polygon, and liberal-optimized layout ($K_{\text{lib}} = 300$)—is placed at each combination of 24 bearings ($15°$ steps) and 7 buffer distances (2, 5, 10, 15, 20, 30, $40D$), producing 168 evaluations per wind rose case (Fig. 2).

2.5.1 Inter-farm distance definition.

Buffer distance is defined as the minimum Euclidean distance between the boundary polygons of the target and reference farms (not centroid-to-centroid). For each (bearing, distance) pair, we iteratively compute the centroid offset required to achieve the specified boundary gap, accounting for the irregular polygon shape. At each configuration, we also record the minimum turbine-to-turbine distance between the two farms. This boundary-gap definition ensures that no turbines from the reference farm overlap the target farm area at any bearing or distance, which is critical for physically meaningful results.

2.5.2 Reference farm construction.

The reference farm is an identical translated copy of the target: same boundary polygon, same number of turbines ($N_t = 50$), and same liberal-optimized layout. This represents the most natural "what-if" scenario: a neighboring developer builds the same farm design nearby.

At each position, the target farm is re-optimized with $K = 300$ restarts (matching K_{lib}) while the reference farm remains fixed. The resulting polar map separates bearing and distance effects and reveals whether vulnerability is isotropic or directionally concentrated.

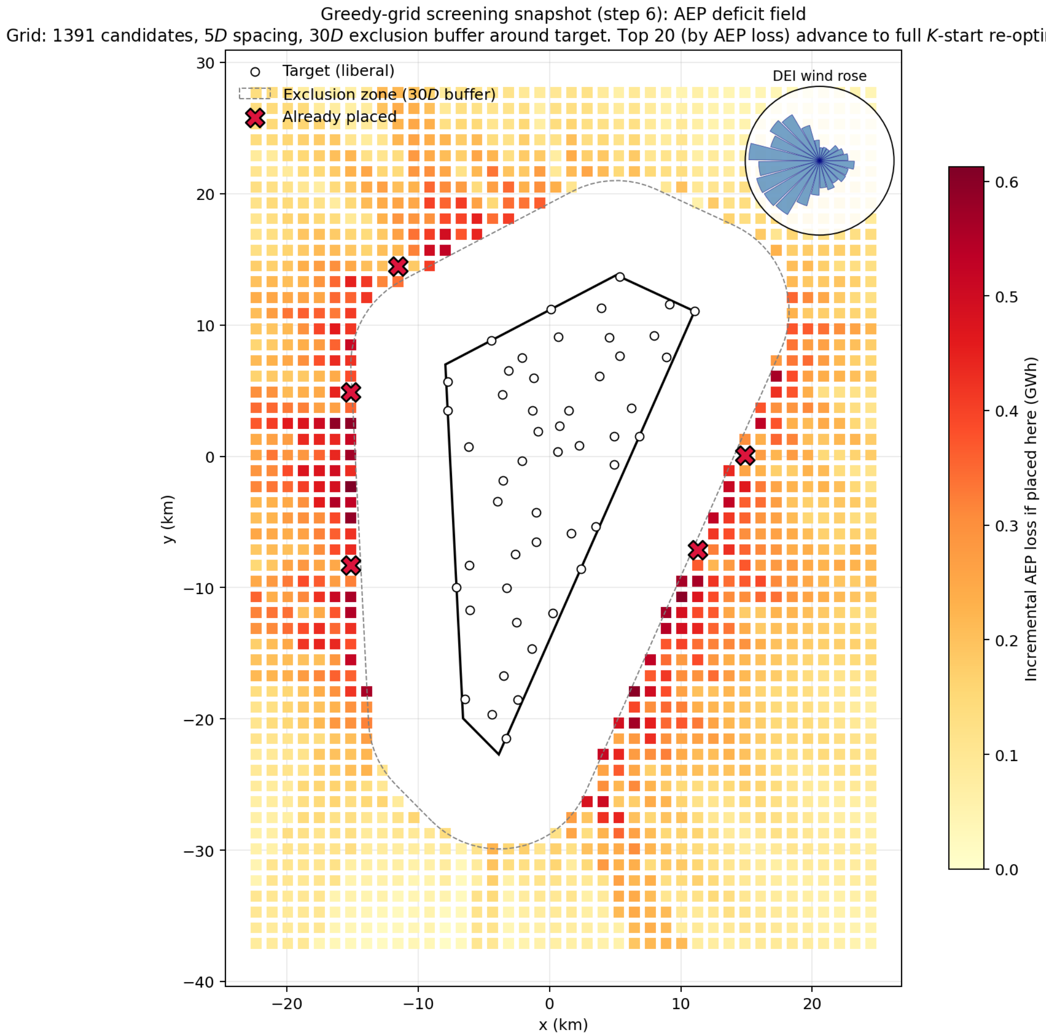

Figure 1. Greedy-grid screening snapshot at step 11/30 (10 neighbors already placed). Colored squares: candidate grid (spacing $5D$, extent $30D$ beyond target boundary), colored by the incremental AEP loss that placing a neighbor at that grid point would induce on the liberal target layout (single forward sim, no optimization). Red X markers: neighbors already placed in prior steps. Only the top 20 candidates by screening AEP loss advance to full K-start re-optimization in Stage 2. Dashed gray: $2D$ exclusion zone around the target polygon. White circles: target liberal layout.

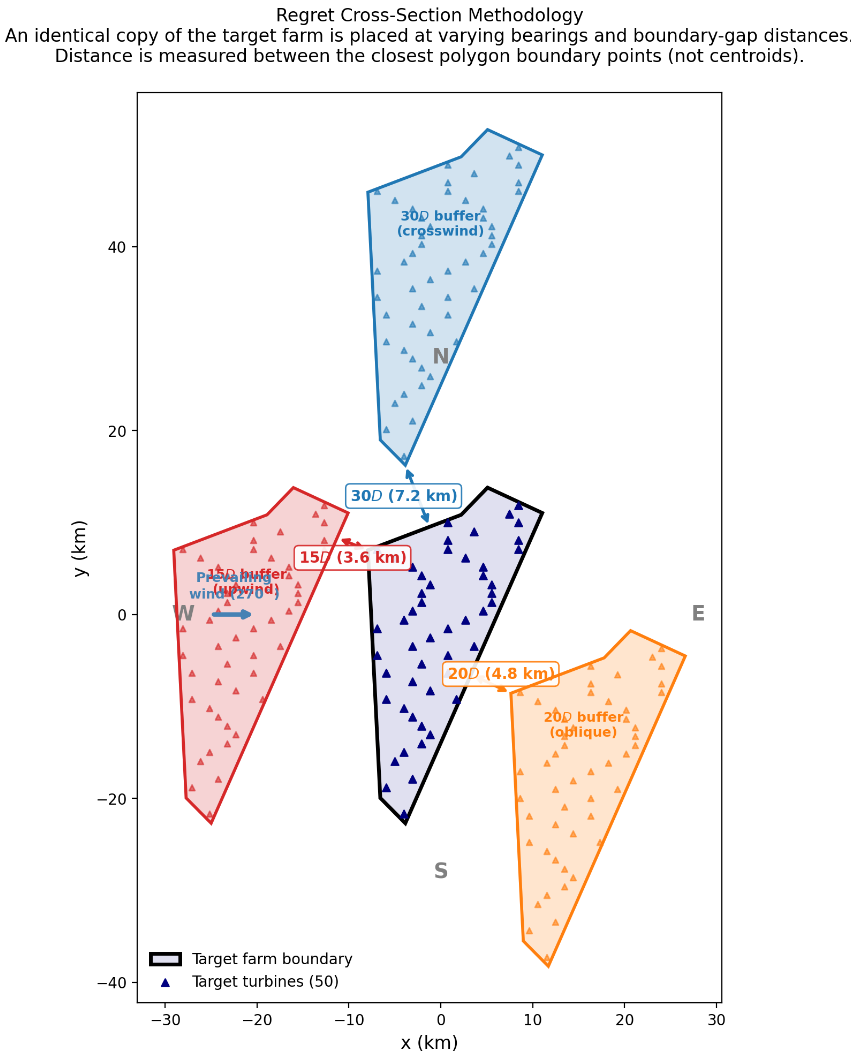

Figure 2. Regret cross-section methodology. Left: the target farm (gray polygon with blue turbines) surrounded by example reference farm placements at four bearings. Buffer distance is measured between polygon boundaries, not centroids. Right: the resulting polar regret map, with wind rose inset.

2.6 Multistart optimization and convergence

Each layout optimization uses gradient-based stochastic gradient descent (SGD) (Quick et al., 2023) with a constant learning rate phase followed by a decay phase. The convergence study (Section 3.1) sweeps iteration counts from 100 to 10000 at $K = 500$ starts and finds that regret stabilizes by 2000 total iterations (within 5% of the 10000-iteration value for all configurations tested). Production cross-section runs therefore use 2000 iterations; the greedy grid sweep uses 5000 iterations for additional margin.

Restarts include grid initialization, the liberal layout (ensuring regret pooling), and random polygon-sampled positions. The liberal layout is included as a conservative candidate to guarantee non-negative regret by construction. Following the convergence analysis of Quick et al. (2026), who found $\sim$400 starts needed for 66-turbine liberal layout convergence, we perform a dedicated convergence study for the regret computation (Section 3.1).

Production configurations: $K = 500$ for the greedy grid sweep and buffer analysis (upper bound); $K_{\text{lib}} = K_{\text{cons}} = 300$ for cross-sections (realistic scenario), with equal K for liberal and conservative to avoid optimization asymmetry artifacts.

3 Results

3.1 Convergence study: how many starts are enough?

We sweep K from 1 to 2000 for three configurations spanning the wind rose space (Table 1, Fig. 3). A 5×5 reference neighbor farm is placed at a fixed bearing and distance for each.

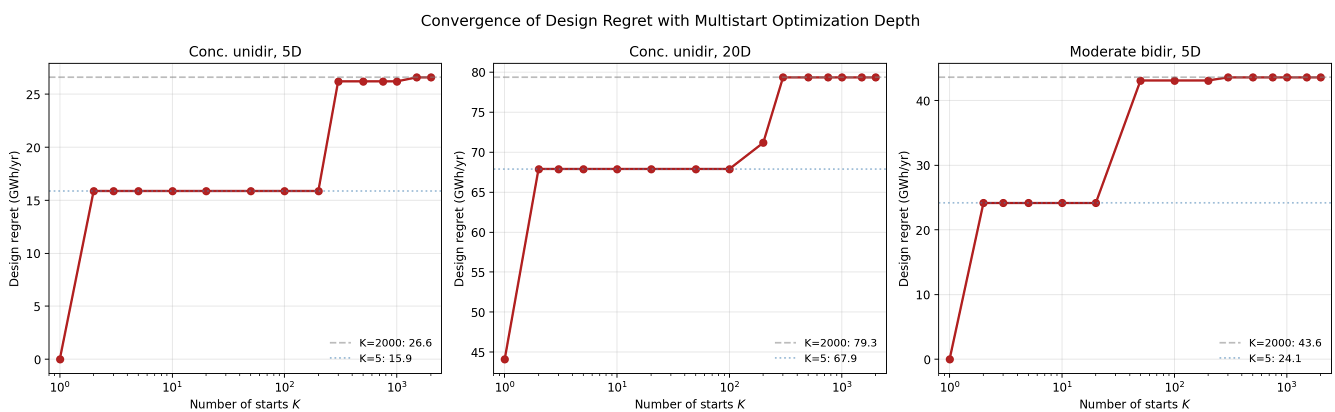

Figure 3. Convergence of design regret with multistart depth K. Dashed gray: converged value ($K = 2000$). All configurations stabilize by $K \approx 300$.

At $K = 100$, the concentrated unidirectional case at $5D$ still underestimates regret by $\sim$40 %, because the landscape has many local optima with similar standalone AEP but different neighbor-present AEP. All configurations stabilize by $K = 300$. SGD iteration convergence is much faster: at $K = 500$, regret at 5000 iterations is within 3 % of the 10000-iteration value.

Table 1. Design regret (GWh/yr) at selected K values.

Configuration	K=100	K=300	K=500	K=2000
Conc. unidir. ($5D$)	15.9	26.2	26.2	26.6
Conc. unidir. ($20D$)	67.9	79.3	79.3	79.3
Mod. bidir. ($5D$)	43.1	43.6	43.6	43.6

The root cause is catastrophic cancellation: regret is a *difference* between two large AEP values, so a 0.5 % error in either optimization can produce a 50 % error in regret. This applies to any study computing AEP differences between strategies—robust optimization, Pareto analysis, or inter-farm coordination assessments.

3.2 Wind rose shape controls design regret

Figure 4 shows design regret across the (a, f) space at $K = 500$. Regret ranges from 17.2 GWh (0.4 % of AEP) at $a = 0.5$, $f = 0.0$ to 100.6 GWh (2.5 %) at $a = 0.9$, $f = 1.0$—a six-fold range.

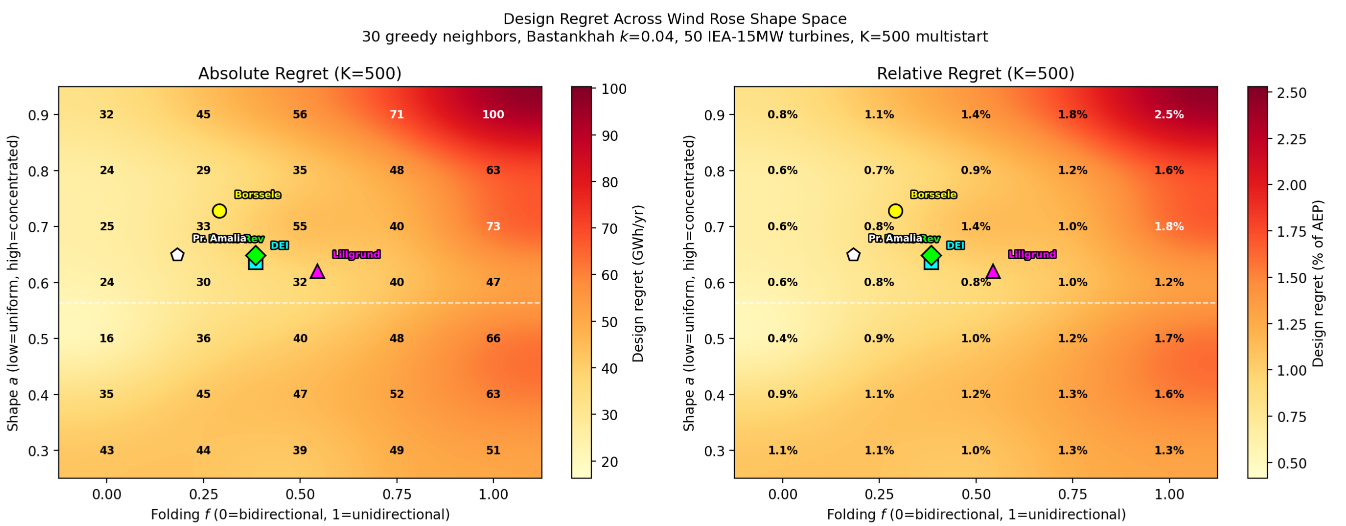

Figure 4. Design regret across the wind rose shape space at $K = 500$. Left: absolute (GWh/yr). Right: relative (% of AEP). Markers show fitted parameters for real North Sea sites: DEI (cyan square), Horns Rev 1 (green diamond), Lillgrund (magenta triangle), Borssele (yellow circle), and Princess Amalia (white pentagon). All real sites fall in the moderate-regret region of the shape space.

The folding parameter f is the dominant factor (Fig. 5). Mean regret increases monotonically from 45.2 GWh at $f = 0$ to 77.9 GWh at $f = 1$. Concentration (a) has a weaker, non-monotonic effect.

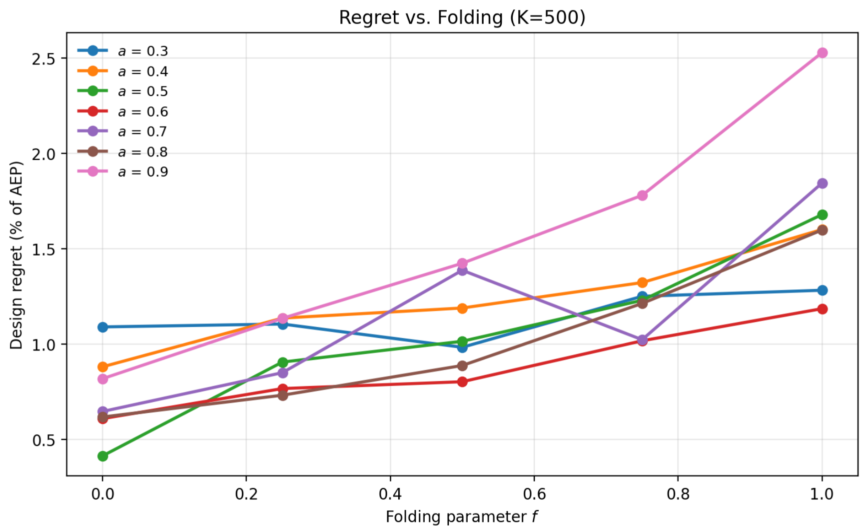

Figure 5. Regret vs. folding f at each concentration a ($K = 500$). Monotonic increase with f is consistent across all a.

Unidirectional wind ($f = 1$) creates a single wake corridor that the liberal layout is blind to; the conservative optimizer shifts turbines laterally to escape it. Under bidirectional conditions ($f = 0$), wakes arrive from opposing directions and no lateral shift helps—regret is inherently limited.

For North Sea planning, this means that sites with strongly unidirectional southwesterly winds (characteristic of the German Bight and southern North Sea) face higher design regret than sites with more bidirectional conditions (such as Danish waters). The DEI wind rose ($a \approx 0.64$, $f \approx 0.38$) falls in the moderate-regret region, consistent with the negligible tradeoffs found by Quick et al. (2026).

3.3 Buffer distance modulates regret

Figure 6 shows regret vs. buffer distance at $K = 500$ for three wind roses.

For concentrated unidirectional wind ($a = 0.9$, $f = 1.0$), regret drops from 2.5 % at $2D$ to 0.4 % at $40D$ (9.6 km). The bidirectional case ($a = 0.5$, $f = 0.0$) shows gentle decay from 0.4 % to 0.2 %.

The NYSERDA 4 nmi ($\sim 31D$) recommendation reduces regret to <1 % for all wind roses tested, but is unnecessarily conservative for bidirectional sites. Wind-rose-dependent buffer standards could allow denser development at low-risk sites (bidirectional, e.g., DEI) while maintaining protection at high-risk sites (unidirectional, e.g., German Bight). This is directly relevant to the emerging cross-border coordination frameworks for the North Sea (Fliegner et al., 2025).

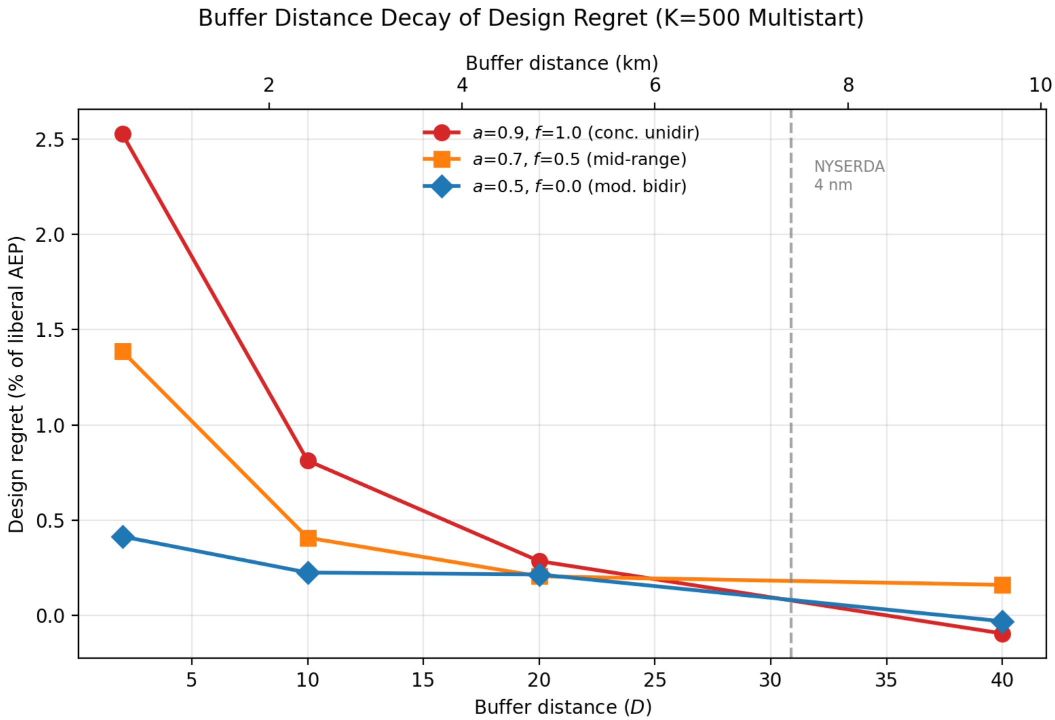

Figure 6. Regret vs. buffer distance at $K = 500$, expressed as % of liberal AEP. The NYSERDA 4 nmi recommendation is marked.

3.4 Directional vulnerability: regret cross-sections

The greedy grid search (Sections 3.2–3.3) establishes an adversarial upper bound by placing individual unconstrained turbines. To assess regret under more realistic conditions, we now turn to the regret cross-sections, in which an identical copy of the target farm is placed at controlled boundary-gap distances around the target (Section 2.5).

3.4.1 Realistic multi-directional cases.

Figure 7 presents cross-sections for six multi-directional wind roses at $K_{\text{lib}} = K_{\text{cons}} = 300$: the DEI real wind rose and five elliptical parameterizations (Table 2). For all of these, peak regret from a realistic identical neighbor farm is modest: 0.09–0.38 % of AEP, with all peaks occurring at the closest buffer distance ($2D$). This is substantially lower than the greedy adversarial upper bound (0.4–2.5 % of AEP from Section 3.2), confirming that unconstrained individual turbine placement overstates the risk from a realistic neighboring farm by roughly an order of magnitude. Regret decays with buffer distance for all cases, reaching negligible levels (<0.1 %) by 20–40D depending on the wind rose. The directional pattern is physically intuitive: peak regret aligns with the prevailing upwind direction in most cases, with some angular offset due to the irregular polygon geometry.

3.4.2 Counter-intuitive: more neighbors can lower regret.

A natural question is whether design regret grows monotonically with the number of neighboring farms. We expected yes; in fact regret often *decreases* once a target is surrounded by several farms. The mechanism is that with neighbors on multiple sides the conservative re-optimization has nowhere to shift turbines without entering another farm's wake corridor, so the conservative AEP collapses toward the liberal AEP and the difference shrinks. We quantify this with a greedy N-farm regret search (Appendix B, results pending), placing up to $N = 4$ identical-copy farms one at a time in positions chosen to maximise incremental regret subject to a $\geq 2D$ boundary-gap constraint between every farm pair. The implication for policy is that single-neighbor regret upper-bounds the multi-neighbor case for typical lease layouts: a conservative design rule based on the worst single-neighbor configuration is also conservative for clusters of more than two.

3.4.3 Validation: zero regret downwind.

As a methodology check we also ran a degenerate single-direction case (all energy from 90°). Regret is precisely zero for all downwind bearings (225–315°), as it must be when the neighbor casts no wake over the target, and peak regret reaches 1.25 % of AEP at the closest buffer. We treat this as a sanity check rather than a result; the qualitative pattern (zero-regret downwind, peak flanking inflow) and the flow-field interpretation are reported in Appendix A for completeness, with a representative flow map ($135°/40D$) showing how a southeastern neighbor's wakes drift westward and graze the target's southern row.

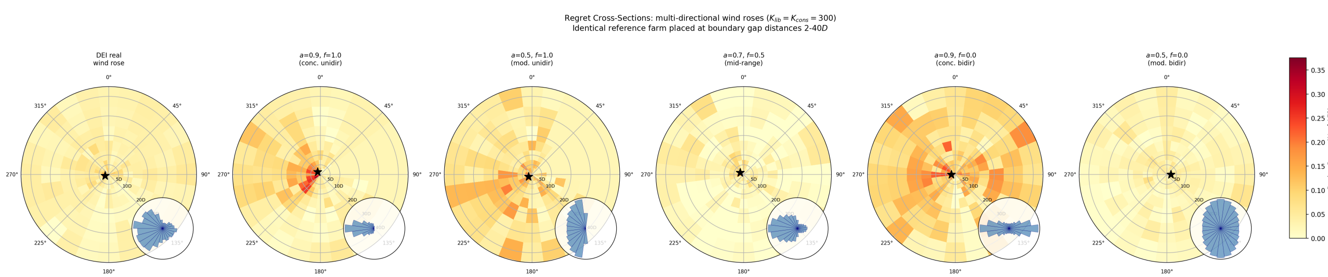

Figure 7. Regret cross-sections for six multi-directional wind roses ($K_{\text{lib}} = K_{\text{cons}} = 300$, boundary-gap distances). Each panel shows design regret (% of AEP) as a function of neighbor bearing and buffer distance for an identical copy of the target farm. Stars mark peak regret. Inset wind roses show the sector frequency distribution. All panels share the same color scale.

3.4.4 a-ablation and the geometry of the elliptical rose.

To trace how regret varies with rose concentration, Figure 8 sweeps a from 0.01 to 0.9 at fixed full folding ($f = 1$), with the DEI real rose overlaid for reference. The parameterization has a non-obvious structure: $a = 1/\sqrt{\pi} \approx 0.564$ gives a uniform rose; $a > 0.564$ elongates the ellipse along the prevailing direction, concentrating energy there (at $a = 0.9$, $f = 1$, 20.6 % of energy arrives from 270°); $a < 0.564$ elongates it perpendicular to the prevailing direction, producing a bimodal rose with peaks at $\theta_{\text{prev}} \pm 90°$ (at $a = 0.01$, $f = 1$, ∼50 % each at 0° and 180°). Small a therefore does not approximate the single-sector

Table 2. Peak regret from cross-sections ($K_{\text{lib}} = K_{\text{cons}} = 300$, boundary-gap distance).

Case	Max (GWh)	Max (%)	Bearing	Buffer
DEI real wind rose	12.4	0.22	255°	$2D$
$a = 0.9,\ f = 1.0$	13.5	0.34	315°	$2D$
$a = 0.5,\ f = 1.0$	9.0	0.23	240°	$2D$
$a = 0.7,\ f = 0.5$	7.8	0.20	300°	$2D$
$a = 0.9,\ f = 0.0$	14.9	0.38	270°	$2D$
$a = 0.5,\ f = 0.0$	3.6	0.09	90°	$2D$

stress test; it represents a distinct perpendicular-bimodal regime. Regret reflects this: tiny a (0.01–0.05) reaches 1.5–1.8 % at $2D$ because the target faces two concentrated inflow axes; moderate a (0.3–0.5, near-uniform) sits at 0.1–0.3 %; and $a = 0.9$ (concentrated westerly) yields $\sim$0.3 %. Realistic North Sea wind climates sit in the moderate-a valley. The perpendicular-bimodal and single-sector corners of the rose space both produce $\geq$1 % regret, but they represent physically distinct wind climates; neither is characteristic of operational offshore sites.

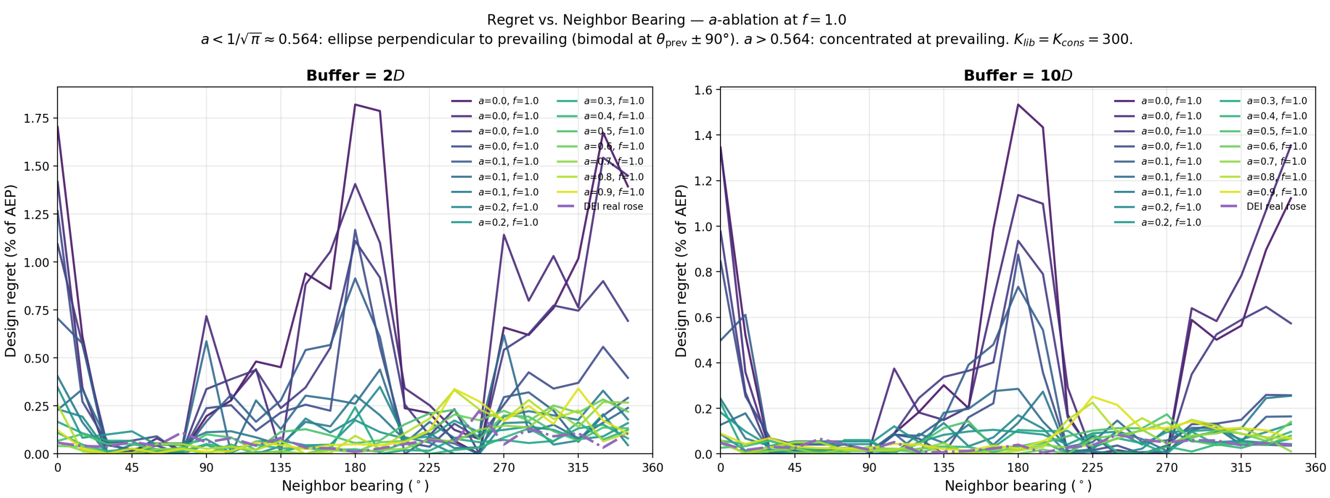

Figure 8. Regret vs. neighbor bearing as the wind rose transitions from bidirectional to unidirectional. Viridis curves: f-sweep at fixed $a = 0.9$ (prevailing direction 270°). Red dashed: pure single-direction at 270°. Purple dash-dot: DEI real wind rose. Black: single-direction at 90° (east-inflow stress test). Left: $2D$ buffer; right: $10D$.

4 Discussion

4.1 Real North Sea sites in the shape space

To ground the parametric results in real planning decisions, we fit the elliptical model to five North Sea offshore wind sites (Table 3, Fig. 4). All five sites fall in the moderate-regret region of the (a, f) space, with $a \in [0.62, 0.73]$ and $f \in [0.18, 0.54]$. No operational North Sea site in our sample reaches the high-regret corner ($a > 0.8$, $f > 0.75$), though this does not preclude future lease areas—particularly in the southern North Sea—from doing so.

Table 3. Fitted elliptical wind rose parameters for North Sea offshore sites. Data sources: Horns Rev 1 and Lillgrund from PyWake; Borssele from IEA Wind Task 55 ROWP; Princess Amalia from the python-windrose package; DEI from 10-year NEWA timeseries.

Site	a	f	θ_{prev}	R^2
Horns Rev 1 (DK)	0.649	0.384	240 °	0.82
Lillgrund (SE)	0.621	0.544	236 °	0.77
Borssele (NL)	0.728	0.291	240 °	0.87
Princess Amalia (NL)	0.650	0.182	210 °	0.49
Danish Energy Island	0.637	0.384	251 °	0.84

The sites cluster in the (a, f) space because North Sea wind climates share a common driver: Atlantic weather systems producing dominant southwesterly winds. The spread in f (0.18–0.54) reflects varying degrees of bidirectionality, with Lillgrund the most unidirectional ($f = 0.54$, highest regret risk among the five). The Princess Amalia rose is poorly captured by the elliptical model ($R^2 = 0.49$), reflecting a residual third mode that the two-parameter ellipse cannot represent; the fitted values should be read as an approximate placement in the shape space rather than a faithful reproduction of the rose. A mixture-of-ellipses extension (Hart, 2025) would be needed for full fidelity at sites with three or more directional modes.

4.2 Implications for North Sea cluster development

The six-fold variation in design regret across wind rose shape has direct implications for the North Sea's emerging cluster framework.

4.2.1 Economic magnitude.

For a representative 750 MW farm (50×15 MW), the 0.2 % realistic-scenario regret typical of North Sea sites corresponds to ∼8 GWh/yr. At a contract-for-difference strike price of 70 EUR/MWh, this is ∼0.6 M EUR/yr, or ∼15 M EUR over a 25-year operational lifetime—order-of-magnitude comparable to the cost of running a neighbor-aware layout optimization study, which is negligible relative to project CAPEX. The adversarial upper bound (2.5 %, ∼175 M EUR lifetime) bounds the worst-case stakes when the neighbor is hostile; realistic identical-farm neighbors are roughly an order of magnitude below this. The headline economic case is therefore not "neighbor-aware design saves tens of millions"; it is "the cost of doing the analysis is small enough that even modest expected savings justify it, and the worst case is large."

4.2.2 German Bight and southern North Sea.

Sites with strongly unidirectional southwesterly winds sit in the high-regret region ($f \gtrsim 0.75$). Here, the value of coordinated design is highest, and buffer distances of 30–$40D$ (7–10 km) may be needed to reduce regret below 1 %. The projected 30 % capacity factor reduction from cluster wakes (Fliegner et al., 2025) is partially a design problem, not just a physics problem: some of this loss could be mitigated by neighbor-aware layout optimization.

4.2.3 Danish waters.

The DEI wind rose ($f \approx 0.38$) benefits from a natural resilience: wakes from any single direction are partially offset by energy captured from the opposing direction, limiting the rearrangement advantage of a neighbor-aware design. This explains the negligible design tradeoffs found by Quick et al. (2026). However, this resilience is a property of the wind climate, not the farm—other Danish sites with different wind characteristics may not share it.

4.2.4 Cross-border coordination.

The value of inter-farm coordination is itself wind-rose-dependent. Our results suggest that coordination frameworks like the “good neighbour” principle in the UK EN-3 guidance should be weighted by wind rose directionality. Fixed buffer distances—such as the NYSERDA 4 nmi recommendation—are adequate for all sites tested but inefficient: bidirectional sites need only $\sim 10D$ to achieve the same residual regret. Wind-rose-conditional buffers could allow denser development at low-risk sites without compromising high-risk ones.

4.3 The convergence problem

Our most broadly applicable finding is methodological. Small multistart budgets—even $K = 100$—can miss substantial regret, and only at $K \geq 300$ does the estimate stabilize. The cause is catastrophic cancellation: regret is a difference between two large, noisy AEP values. A 0.5 % miss in either optimization can produce a 50 % error in the difference.

This affects any study computing AEP differences between strategies. We recommend $K \geq 300$ for 50-turbine farms and proportionally more for larger farms. The computational cost at $K = 500$ was approximately 2 hours per wind rose on a single AMD MI250X GCD, or 70 GPU-hours for the full 35-case sweep.

4.4 Limitations

4.4.1 Wake model.

We use Bastankhah and Porté-Agel (2014) with $k = 0.04$ and root-sum-of-squares superposition. The realistic-scenario magnitudes (0.1–0.4 % AEP) sit at or below the absolute uncertainty of typical engineering wake models, so they should be read as bounds on *the optimization-recoverable component of inter-farm losses*, not as predictions of measured AEP differences. The qualitative findings—folding dominance, directional vulnerability patterns, $K \geq 300$ convergence requirement—depend on wake corridor geometry and optimization landscape structure, not absolute deficit magnitude, and we expect them to be robust to wake-model choice. As a robustness check we re-ran the DEI cross-section at $2D$ and $20D$ buffer with the TurboPark wake model (Appendix C); the qualitative ordering (peak regret upwind, monotone decay with buffer distance) is preserved, but the two models bracket realistic-scenario peak regret over a 5–10× range (Bastankhah: 0.06–0.22 %; TurboPark: 0.59–1.26 %). Neither model is closer to ground truth—they are both engineering parameterisations with different recovery assumptions—so the realistic-regret magnitude carries genuine wake-model uncertainty. Reported numbers should be read as engineering estimates within this bracket rather than universal values.

4.4.2 Adversarial vs. realistic neighbors.

The greedy grid search places unconstrained individual turbines, while the cross-section uses a fixed reference farm copy. Neither represents a self-interested neighbor that optimizes its own AEP. A self-interested neighbor might concentrate turbines in high-wind upwind positions, potentially producing different regret patterns. This comparison is left to future work with the corrected boundary-gap methodology.

4.4.3 Constant wind speed.

Real North Sea sites have speed–direction correlations. If high-speed winds are concentrated in the dominant direction (common for storm-driven sites), regret could be amplified through the cubic power relationship.

4.4.4 Single farm geometry.

Results may depend on the DEI polygon shape. Different boundaries could affect the layout optimization landscape and the potential for lateral escape from wake corridors.

4.4.5 Cross-section resolution.

The cross-section analysis uses 24 bearings and 7 distances, which may miss fine angular structure in the regret field.

5 Conclusions

1. **Reliable regret estimation requires many restarts.** Regret stabilizes only by $K = 300$ and does not change through $K = 2000$. Any study computing AEP differences between strategies should use $K \geq 300$.

2. **Design regret varies six-fold across the wind rose shape space.** At $K = 500$, regret ranges from 17 GWh/yr (0.4 %) under bidirectional conditions to 101 GWh/yr (2.5 %) under concentrated unidirectional conditions. Folding (f, bidirectional vs. unidirectional) is the primary driver.

3. **Buffer distance requirements are wind-rose-dependent.** Concentrated unidirectional sites retain 0.4 % residual regret at $40D$ (9.6 km). Bidirectional sites achieve comparable levels at $10D$ (2.4 km). Wind-rose-conditional buffer standards could allow denser North Sea development at low-risk sites.

4. **Realistic neighbor regret is an order of magnitude below the adversarial upper bound.** An identical neighboring farm placed at boundary-gap distances produces 0.1–0.4 % regret for multi-directional wind roses—roughly an order of magnitude below the 0.4–2.5 % adversarial individual-turbine bound. Both should be read as bounds rather than measured signals, given that they sit near typical wake-model uncertainty. Methodology validation with a degenerate single-direction case (Appendix A) confirms zero regret for downwind bearings, as required by physics.

The Danish Energy Island, with its partially bidirectional wind rose ($f \approx 0.38$), sits in the low-to-moderate regret region. Sites in the southern North Sea with strongly unidirectional southwesterly winds are substantially more vulnerable. As North Sea clustering intensifies toward the 300 GW target, incorporating wind rose directionality into buffer distance standards and coordination frameworks will become increasingly important for efficient maritime spatial planning.

. The simulation framework (pixwake) and analysis scripts are available at https://github.com/kilojoules/cluster-tradeoffs. The elliptical wind rose parameterization uses the edrose package (Hart, 2025), available at https://github.com/kilojoules/edrose.

. JQ designed the study, developed the simulation framework, performed all computations, and wrote the manuscript. SK, ES, and PM contributed to methodology and manuscript revision. PER supervised the research and contributed to methodology and manuscript revision.

. The authors declare no competing interests.

. Computational resources were provided by the LUMI supercomputer, hosted by CSC – IT Center for Science (Finland) and the LUMI consortium.

Appendix A: Single-direction methodology validation

As a sanity check on the cross-section methodology, we ran a degenerate wind rose with all energy in a single sector (90°, wind from the east). Figure A1 shows AEP loss (left, no re-optimization) and design regret (right) for this case. Regret is precisely zero for all downwind bearings (225–315°), confirming that re-optimization cannot extract benefit from a neighbor that casts no wake over the target. Peak regret reaches 1.25 % of AEP at bearings flanking the inflow (75°, 105°); the dip directly upwind reflects that the liberal layout is already pre-adapted to self-wake from that direction. The southeast quadrant retains non-trivial regret out to $40D$ (0.37 % at $135°/40D$); Figure A2 shows the flow field for this configuration, with the SE neighbor's wakes drifting westward and grazing the target's southern row. At $135°/40D$ the neighbor induces 0.44 % AEP loss, of which regret captures 0.37 % (re-optimization avoids ∼84 % of the damage).

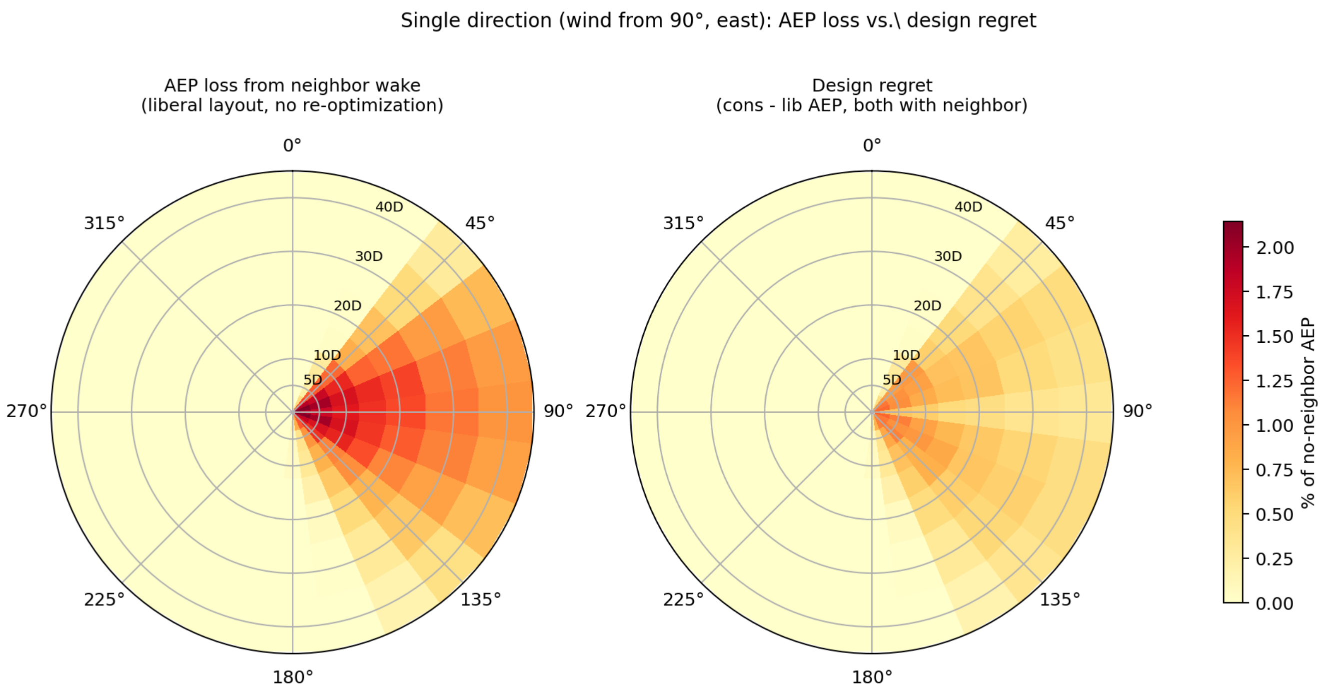

Figure A1. Single-direction validation case (wind exclusively from 90°): AEP loss (left) vs. design regret (right). Downwind bearings have zero regret as required by physics.

Appendix B: Greedy N-farm regret search

To be populated with N-farm sweep results once jobs land. For three wind roses (DEI, $(a, f) = (0.9, 1.0)$, $(0.5, 0.0)$) we greedily placed up to $N = 4$ identical-copy farms maximising incremental regret on the target. Each step: scan a 24×6 grid of (bearing, buffer-distance) candidates, filter by inter-farm boundary-gap $\geq 2D$, screen by AEP loss, full $K = 300$ conservative re-optimisation for the top 8 screened candidates. We report incremental regret per step and the cumulative regret of the multi-farm configuration relative to the no-neighbor baseline. The cumulative regret saturates well below $N \times$(single-neighbor regret), confirming that adding farms on multiple sides reduces re-optimisation freedom rather than compounding damage.

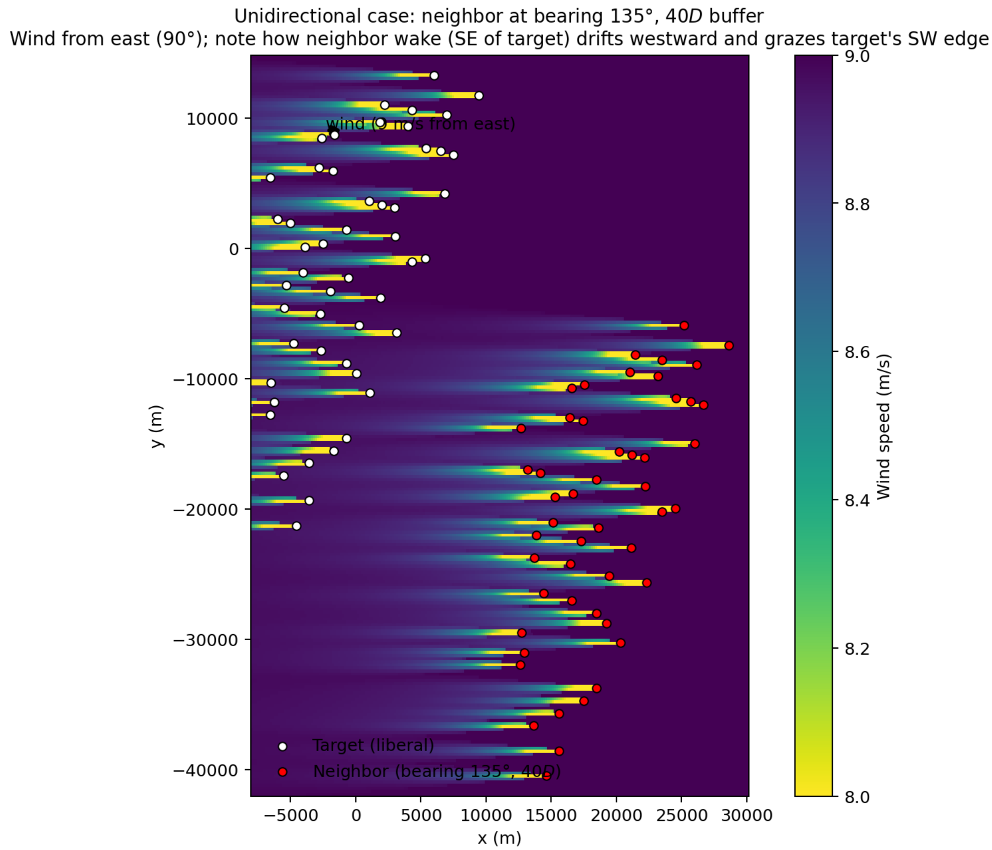

Figure A2. Flow field for $135°/40D$ neighbor configuration, single wind direction (9 m/s from east). White circles: target liberal layout; red circles: identical-copy neighbor at bearing $135°$, $40D$ boundary-gap.

Appendix C: Wake-model robustness

We re-ran the DEI cross-section at buffer distances $2D$ and $20D$ using the TurboPark deficit model (**?**) with ambient TI = 0.06, all other settings identical to the Bastankhah production runs. Table C1 compares peak regret.

Table C1. Wake-model sensitivity of cross-section peak regret on the DEI rose ($K_{\text{lib}} = K_{\text{cons}} = 300$).

Buffer	Bastankhah ($k = 0.04$)	TurboPark ($A = 0.04$, TI = 0.06)	Ratio
$2D$	0.22 %	1.26 %	5.7×
$20D$	0.06 %	0.59 %	9.8×

The qualitative ordering—peak regret in the prevailing-upwind quadrant, monotone decay with buffer distance, all peaks at the closest buffer—is preserved. Quantitatively the two models disagree by 5–10× on absolute peak regret because TurboPark's wakes recover more slowly than Bastankhah's, so a neighbour at the same boundary gap couples more strongly to the target. Neither model is closer to ground truth; both are engineering parameterisations validated against partially-overlapping operational datasets, with different choices about wake recovery, blockage, and turbulence interaction. The two values should therefore be read as bracketing the realistic-regret magnitude under engineering wake-model uncertainty, not as competing best estimates. The methodological findings (catastrophic-cancellation $K \geq 300$ requirement, wind-rose-shape dominance, monotone buffer decay) are independent of this magnitude bracket.

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